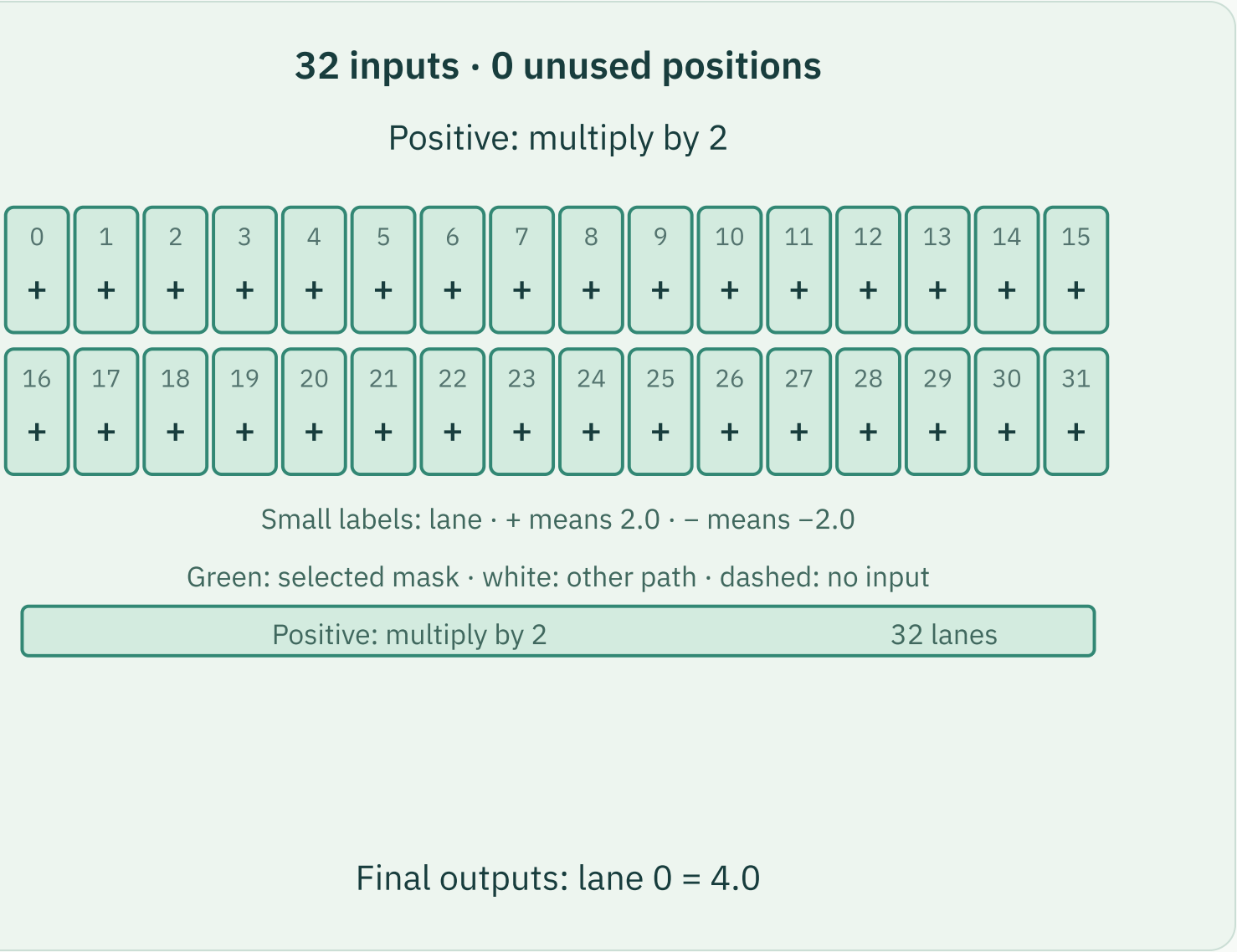
1 / 8

A mask names participating lanes.

All 32 positive inputs use the same calculation.

> Empty arithmetic paths are skipped.



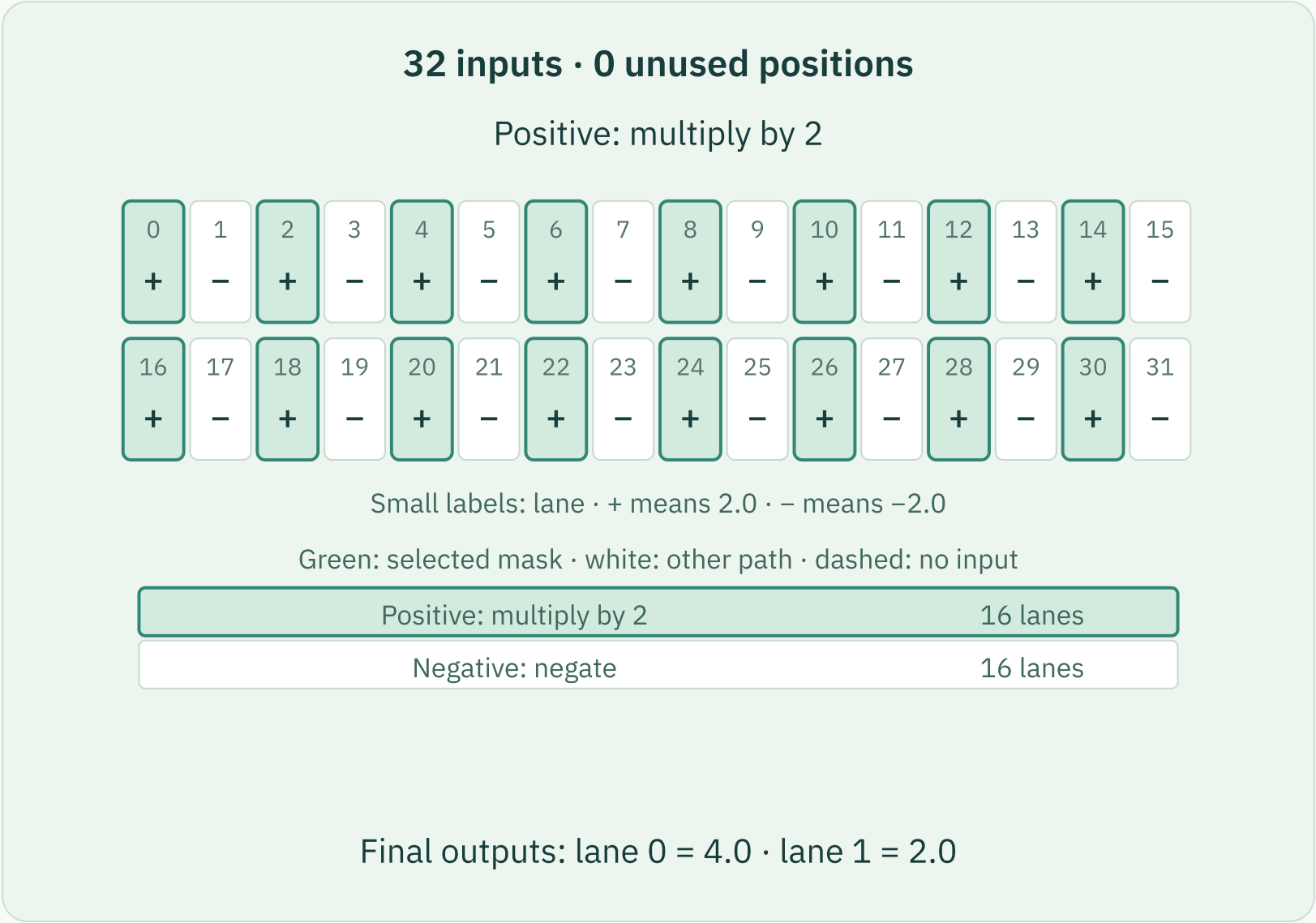
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Mixed signs need different masks.

The positive operation uses only the positive lanes.

> Other-path lanes still have inputs.



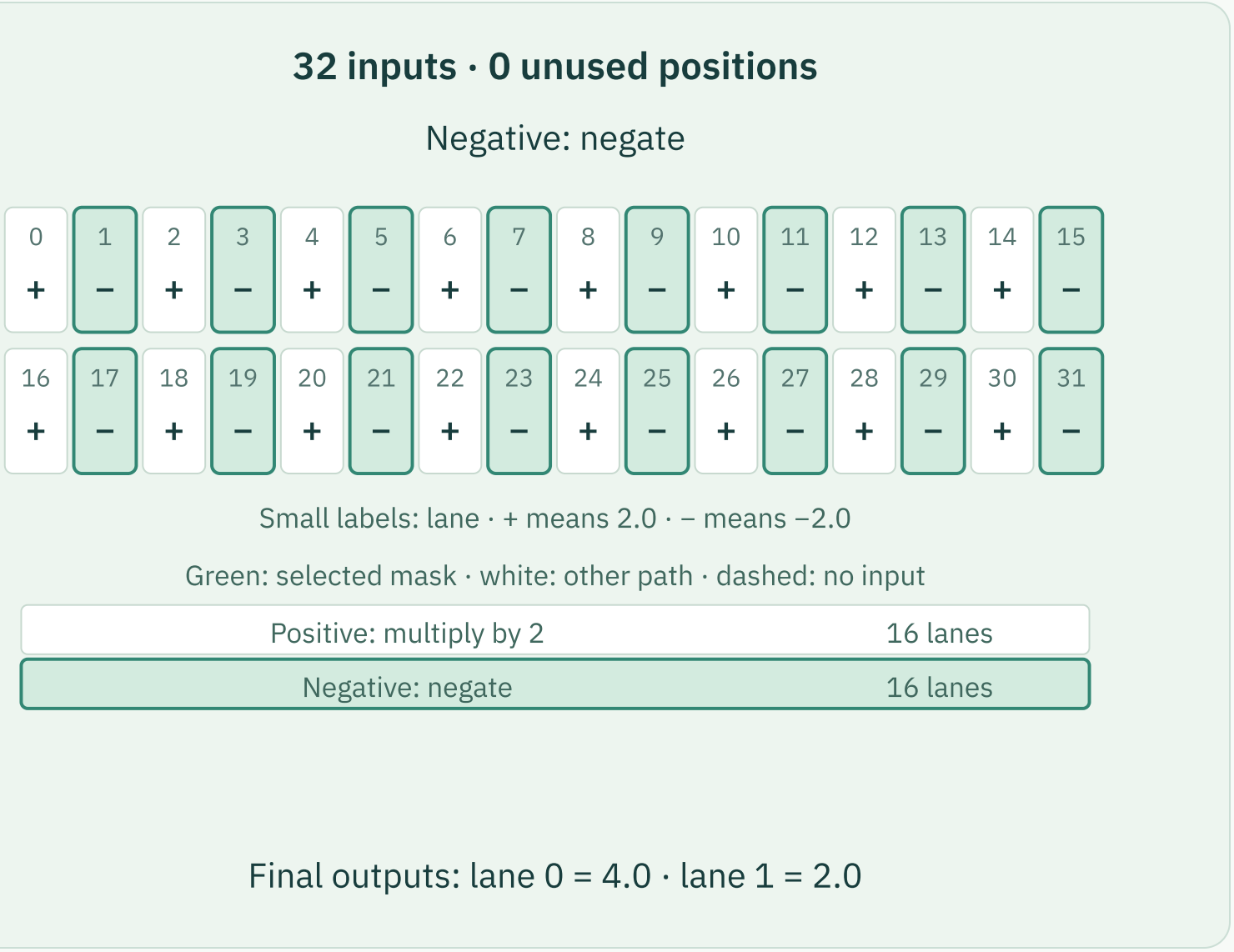
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Other-path lanes participate later.

The negative operation selects the odd lanes.

> The positive-first order is chosen for this model.



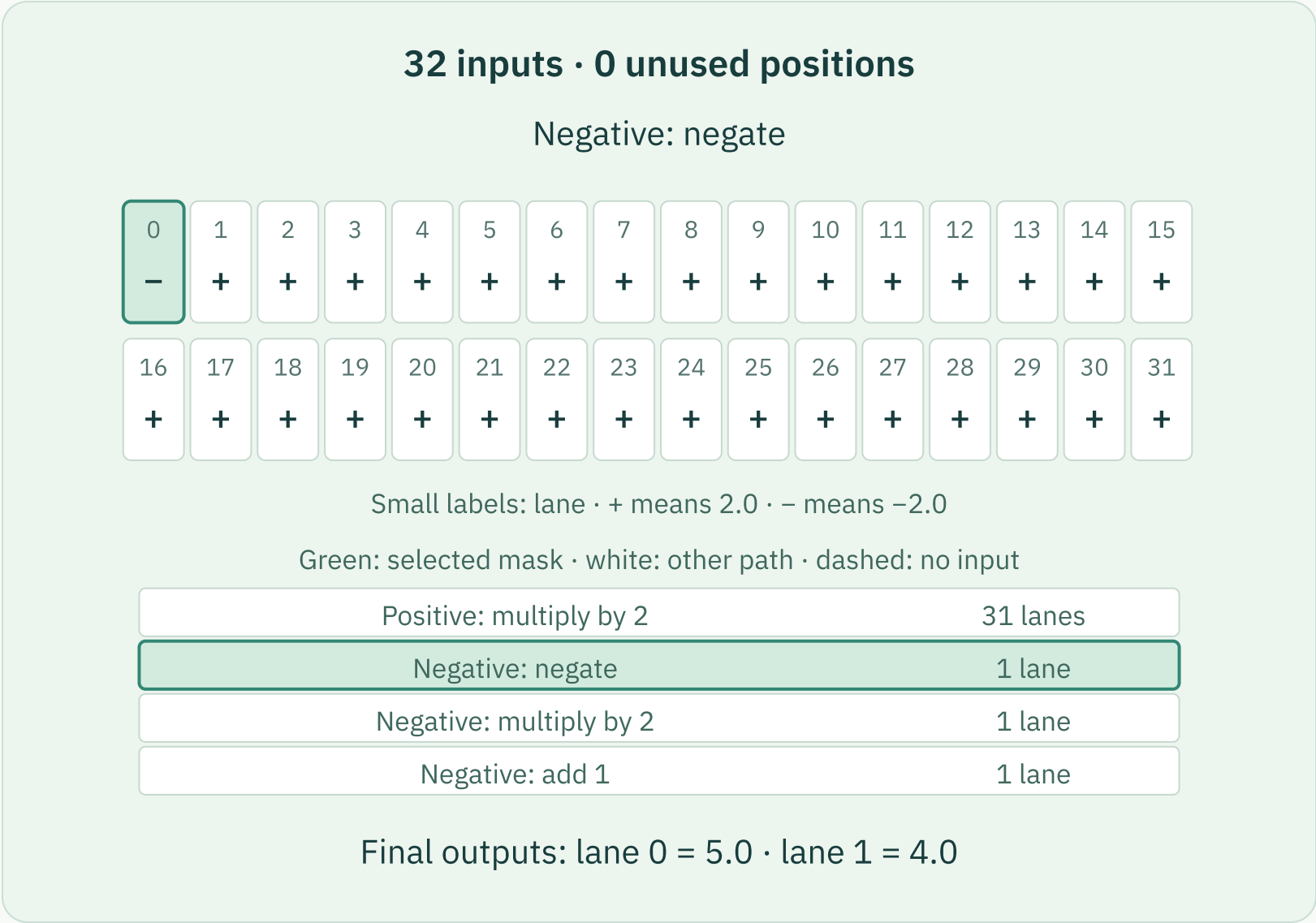
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A rare path can still contain several operations.

Only lane zero is negative, but its path is non-empty.

> Only the listed arithmetic is counted.



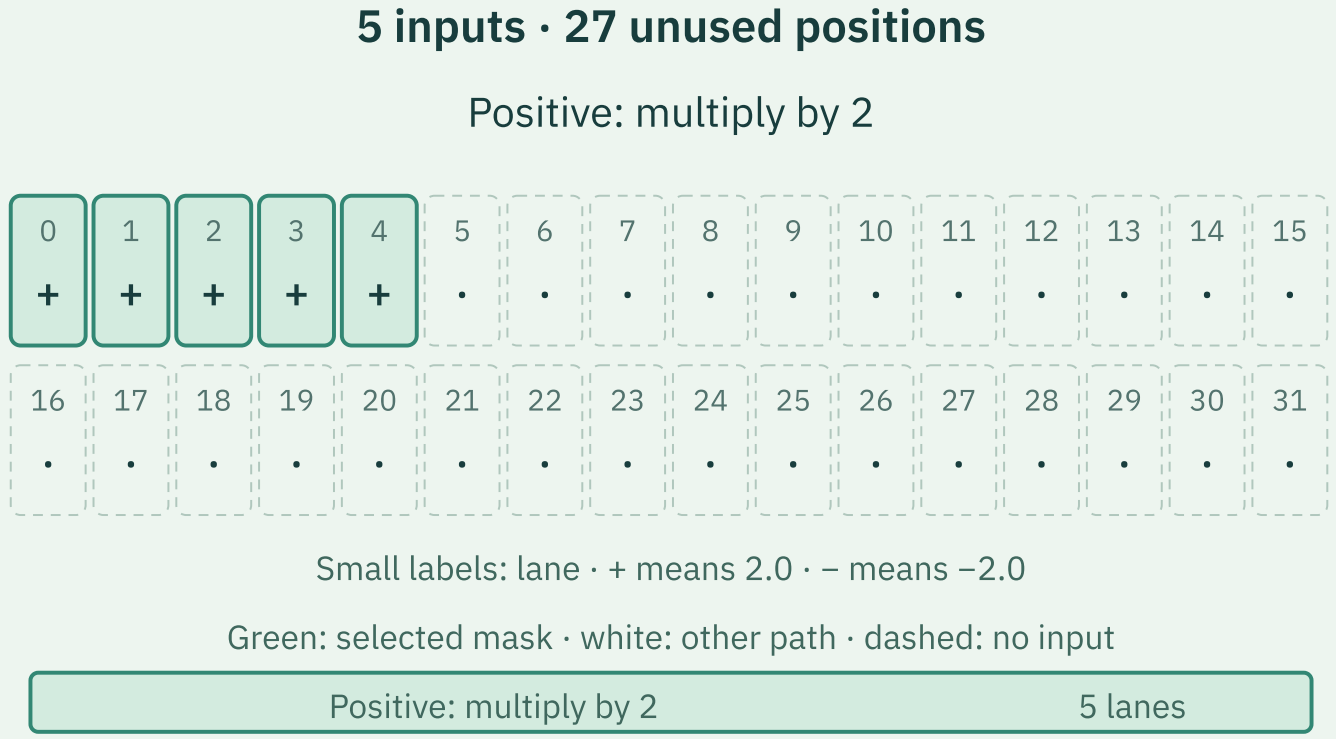
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Unused positions have no input.

Five positive inputs need just one path.

> One non-empty path: low participation without divergence.



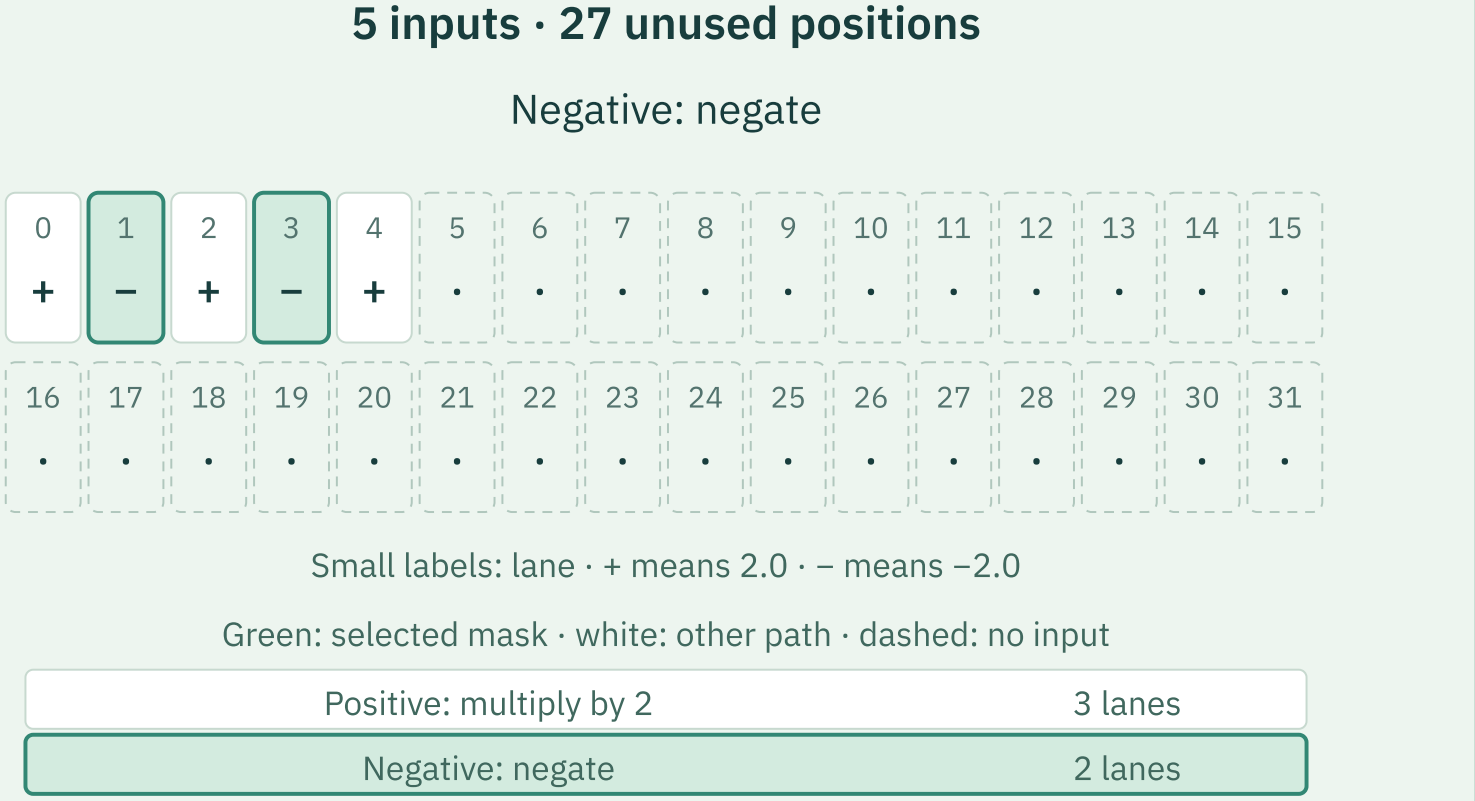
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A small input can show both effects.

Five alternating inputs use two masks and leave a tail.

> The selected mask and the unused tail are different sets.



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Pause: one negative input after sixteen positives.

17 inputs. The negative path has three operations.

> Use the input signs and skip only empty paths.

17 inputs · 15 unused positions

Predict the masks and arithmetic trace

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
−	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·

Small labels: lane · + means 2.0 · − means −2.0

Positive: input × 2

Negative: negate → × 2 → + 1

Predict each non-empty mask and the operation count.

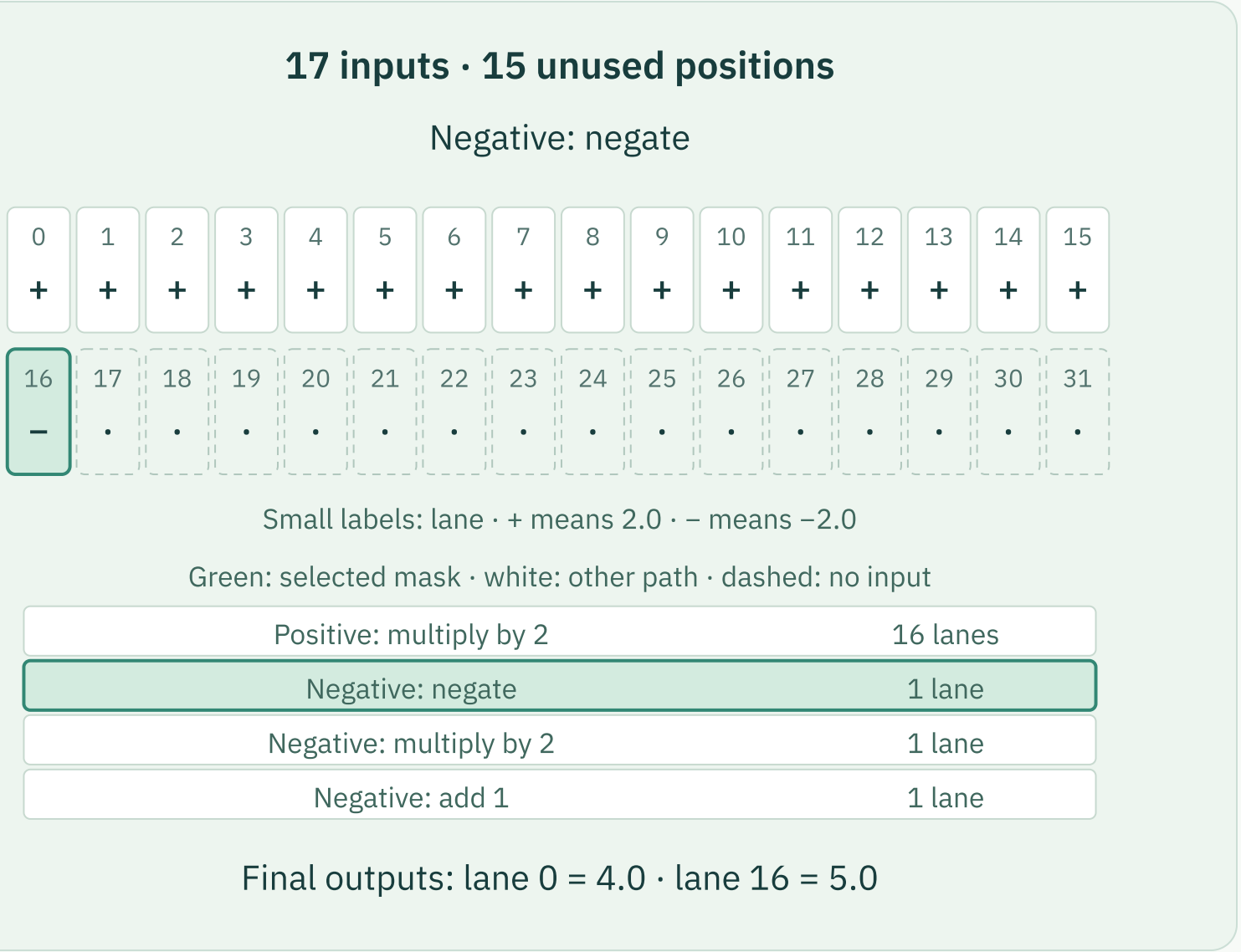
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One negative lane keeps its whole path.

Four arithmetic operations; 15 positions stay unused.

> Masks explain participation; these are not compiled timings.



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